Lab 05: Routing Between Different Networks

Static Routes Across Multiple Routers

Lab #5	**Topic:** Inter-Network Routing	**Tool:** Cisco Packet Tracer	**Cert:** CompTIA Network+

1. Lab Objectives

- Configure two routers connected via a WAN link (serial or Ethernet).
- Set up static routes so PCs on different networks can reach each other.
- Understand how routing tables are built and consulted.
- Distinguish between directly connected routes and static routes.
- Relate to Network+ objective: 1.4 – Explain routing concepts (static vs dynamic routing).

2. Network Topology

Three networks are used. LAN A (10.0.1.0/24) is behind R1, LAN B (10.0.2.0/24) is behind R2, and the WAN link between R1 and R2 uses the 10.0.0.0/30 subnet.

Device	Interface	IP Address	Network
R1	Fa0/0 (LAN A)	10.0.1.1	10.0.1.0/24
R1	Fa0/1 (WAN)	10.0.0.1	10.0.0.0/30
R2	Fa0/0 (WAN)	10.0.0.2	10.0.0.0/30
R2	Fa0/1 (LAN B)	10.0.2.1	10.0.2.0/24
PC0	Fa0 (LAN A)	10.0.1.10	GW: 10.0.1.1
PC1	Fa0 (LAN B)	10.0.2.10	GW: 10.0.2.1

3. Step-by-Step Configuration

3.1 Router R1

```
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 10.0.1.1 255.255.255.0
R1(config-if)# no shutdown
```

```
R1(config-if)# exit

R1(config)# interface FastEthernet0/1
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

! Static route to LAN B via R2
R1(config)# ip route 10.0.2.0 255.255.255.0 10.0.0.2
R1(config)# exit
R1# write memory
```

3.2 Router R2

```
R2(config)# interface FastEthernet0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit

R2(config)# interface FastEthernet0/1
R2(config-if)# ip address 10.0.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit

! Static route to LAN A via R1
R2(config)# ip route 10.0.1.0 255.255.255.0 10.0.0.1
R2(config)# exit
R2# write memory
```

4. Verification

```
PC0> ping 10.0.2.10         ! PC0 → PC1 across both routers
R1# show ip route           ! See connected + static routes
R2# show ip route           ! Verify return path exists
R1# traceroute 10.0.2.10    ! Shows hop-by-hop path
```

5. Key Takeaways

- Static routes must be configured in BOTH directions — routing is not automatically bidirectional.
- A /30 mask (255.255.255.252) is ideal for point-to-point WAN links — only 2 usable host IPs.
- The routing table shows: C = Connected, S = Static, O = OSPF, R = RIP.
- If a static route is missing on one router, ping will fail on the return path (asymmetric routing).
- For larger networks, dynamic routing protocols (OSPF, EIGRP) replace static routes to automate this process.